DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Summer Examination -2024

Course: B. Tech.

Branch : Computer Engineering and Allied

Semester :IV

Subject Code & Name: Design and Analysis Of Algorithms(BTCOC401)

Max Marks: 60 **Date:12/06/2024** **Duration: 3 Hrs.**

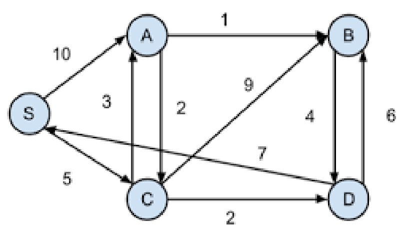
Instructions to the Students:

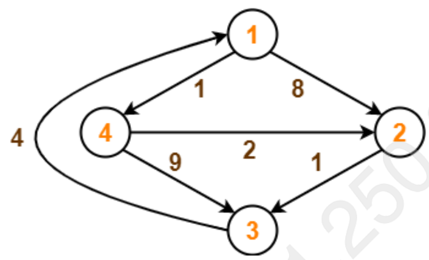
1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	**Solve the following recurrences using Master Theorem.** **(i)** $T(n) = 16T(n/4) + n$. **(ii)** $T(n) = T(2n/3) + 1$ **(iii)** $T(n) = 3T(n/4) + n \log n$	**Apply**	**6**
B)	**What are different asymptotic notations? Explain with neat graphs. If f(n) = sqrt (n) and g(n)=n then prove that f(n) is Og(n).**	**Analyze**	**6**
C)	**What are the different properties of algorithm ? Which factors affect runtime of an algorithm ?**	**Understand**	**6**
Q.2	**Solve Any Two of the following.**		
A)	**Apply Strassen's matrix multiplication algorithm on following matrices.** $\left\{\begin{matrix} 2 & 3 \\ 6 & 8 \end{matrix}\right\} \quad \left\{\begin{matrix} 1 & 2 \\ 3 & 6 \end{matrix}\right\}$	**Apply**	**6**
B)	**What is best case and worst case time complexity for Merge Sort ? Apply Merge Sort on following array.** **A = { 13, 4, 22, 1,16, 9,0,2}**	**Apply**	**6**
C)	**Distinguish between Divide and Conquer and Dynamic programming approach .**	**Understand**	**6**
Q. 3	**Solve Any Two of the following.**		
A)	**Apply Huffman Coding algorithm on following data. Write the code for each character using Huffman Coding algorithm.**	**Apply**	**6**

Character	**a**	**b**	**c**	**d**	**e**	**f**
Frequency	**45**	**13**	**12**	**16**	**9**	**5**

<table>
<tr><td>B)</td><td>Apply greedy approach to solve following fractional knapsack problem if capacity of knapsack is 60 kg. Calculate maximum profit.
<table>
<tr><th>Item</th><th>Weight(in KG)</th><th>Value (Profit)</th></tr>
<tr><td>1</td><td>5</td><td>30</td></tr>
<tr><td>2</td><td>10</td><td>40</td></tr>
<tr><td>3</td><td>15</td><td>45</td></tr>
<tr><td>4</td><td>22</td><td>77</td></tr>
<tr><td>5</td><td>25</td><td>90</td></tr>
</table></td><td>Apply</td><td>6</td></tr>
<tr><td>C)</td><td>Caculate shortest path from node S to all nodes for following graph using Dijkstra’s algorithm.


</td><td>Apply</td><td>6</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>Q.4</td><td>Solve Any Two of the following.</td><td></td><td></td></tr>
<tr><td>A)</td><td>Apply Floyd Warshall’s algorithm on following graph.


</td><td>Apply</td><td>6</td></tr>
<tr><td>B)</td><td>Determine Longest Common Subsequence between X and Y if X=< x m j y a u z> and Y= <m z j a w x u>.</td><td>Apply</td><td>6</td></tr>
<tr><td>C)</td><td>Explain the following terms.
(i) P -problem (ii) NP problem (iii)NP complete problem</td><td>Understand</td><td>6</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>Q. 5</td><td>Solve Any Two of the following.</td><td></td><td></td></tr>
<tr><td>A)</td><td>How 4 –Queens problem is solved by backtracking approach ? Draw state space tree for 4- Queens problem?</td><td>Analyze</td><td>6</td></tr>
<tr><td>B)</td><td>Solve the following travelling salesman problem using branch and bound approach.</td><td>Apply</td><td>6</td></tr>
</table>

	$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array}\begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} \infty & 20 & 30 & 10 & 11 \\ 15 & \infty & 30 & 10 & 11 \\ 3 & 5 & \infty & 2 & 4 \\ 19 & 6 & 18 & \infty & 3 \\ 16 & 4 & 7 & 16 & \infty \end{bmatrix} \end{array}$$		
C)	**Compare and contrast backtracking and branch and bound approach.**	**Understand**	**6**
	***** End *****		